SEC Analysis — Supporting Material

Per-Construct Chromatograms and Peak Tables

Date: April 2026
Analyst: BioClaw
Project: Appendix to SEC Analysis Report

A. Individual Chromatogram Grid

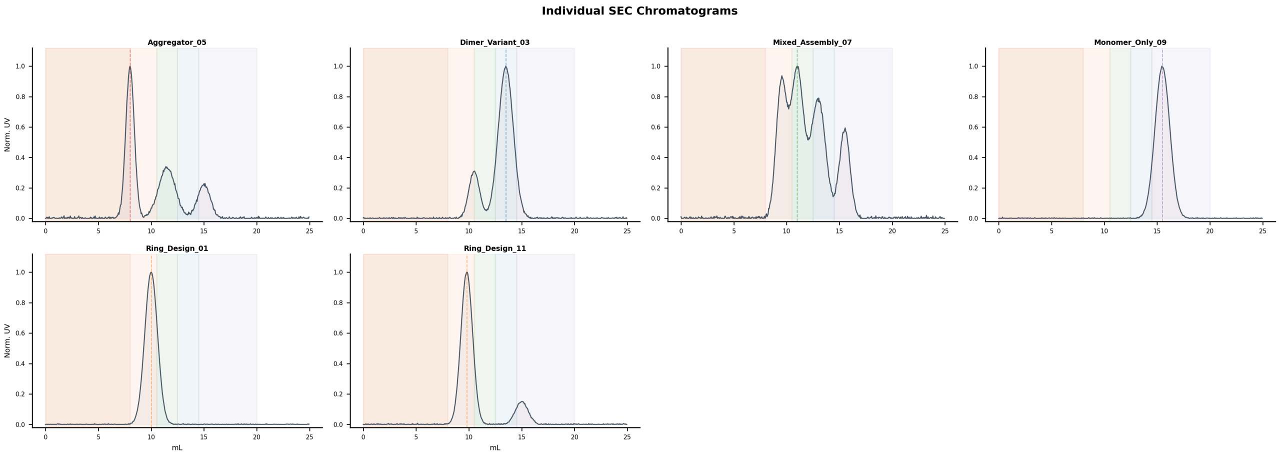

Figure 1: Individual SEC chromatograms for all 6 constructs. Elution zones shaded; dashed line marks dominant peak.

B. Zone Fraction Analysis

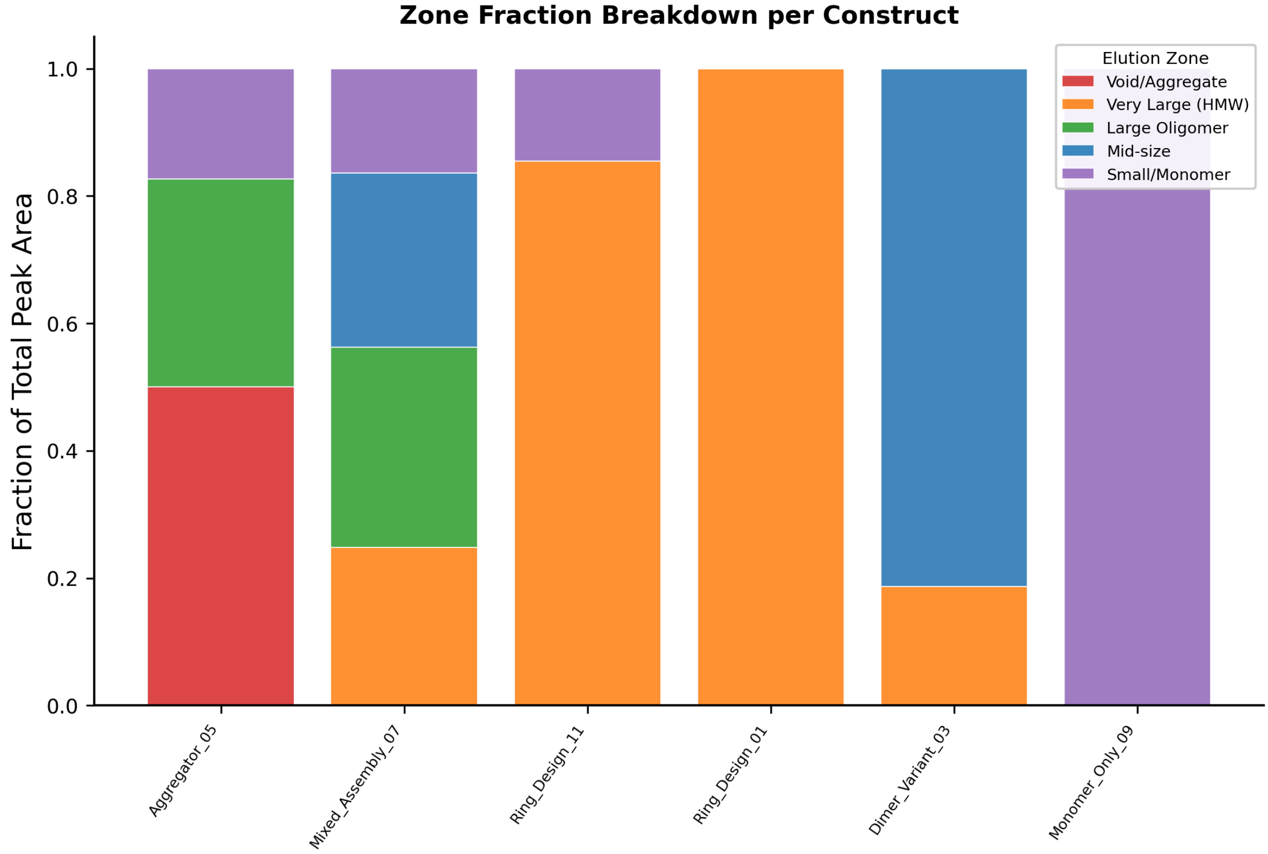

Figure 2: Fraction of total peak area in each elution zone.

C. Per-Construct Analysis

Monomer_Only_09

#	VE (ML)	HEIGHT (MAU)	FWHM (ML)	AREA %	ZONE
1	15.50	95.0	1.65	100.0%	Small / Monomer

Dominant peak at 15.5 mL (Small / Monomer zone), 100% of total area, FWHM = 1.65 mL. Highly monodisperse profile.

Recommendation: Monomer control; SEC-MALS for MW

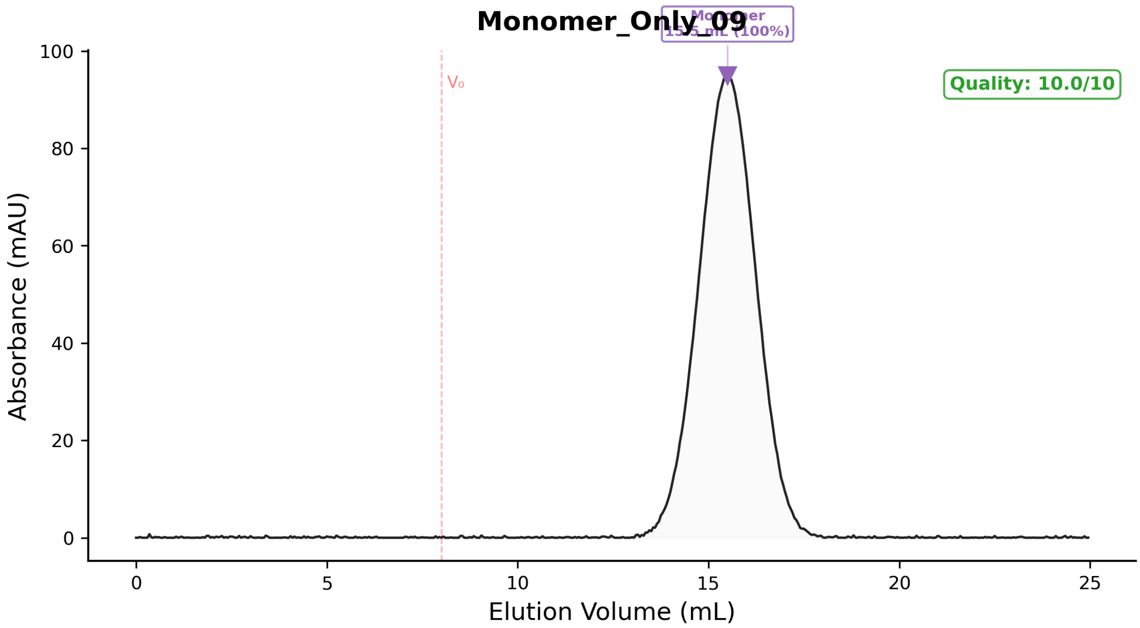

Figure 3: Annotated SEC chromatogram — Monomer_Only_09.

Ring_Design_01

#	VE (ML)	HEIGHT (MAU)	FWHM (ML)	AREA %	ZONE
1	10.00	119.8	1.42	100.0%	Very Large (HMW)

Dominant peak at 10.0 mL (Very Large (HMW) zone), 100% of total area, FWHM = 1.42 mL. Highly monodisperse profile. Elution in the HMW zone is consistent with higher-order oligomeric assembly.

Recommendation: Priority: cryo-EM / native MS

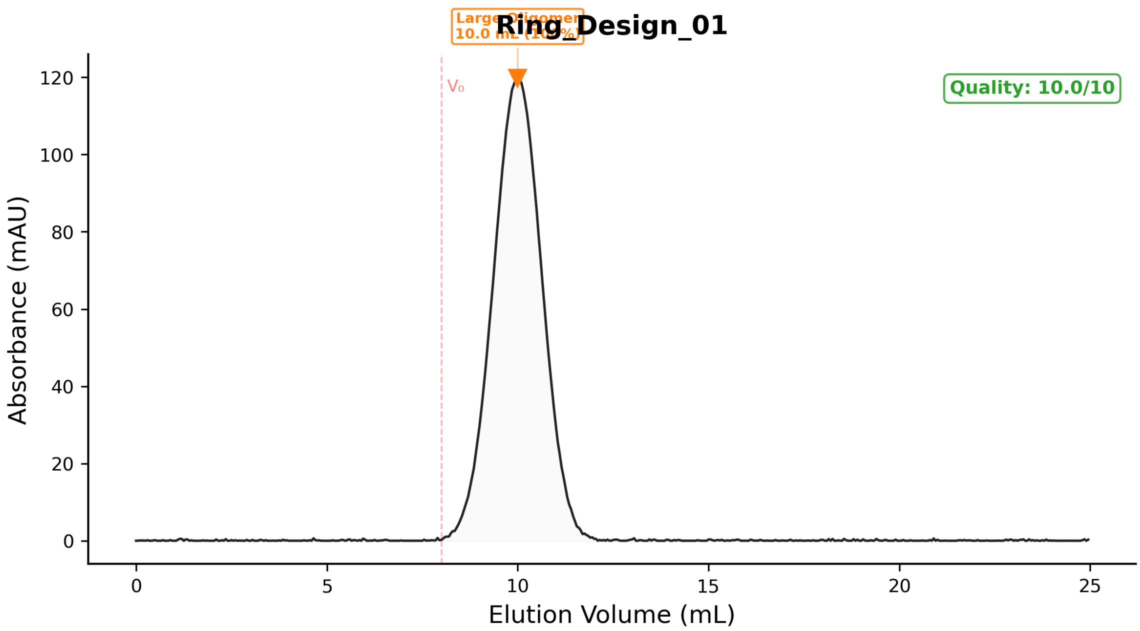

Figure 4: Annotated SEC chromatogram — Ring_Design_01.

Ring_Design_11

#	VE (ML)	HEIGHT (MAU)	FWHM (ML)	AREA %	ZONE
1	9.80	99.7	1.30	85.5%	Very Large (HMW)
2	15.00	14.9	1.42	14.5%	Small / Monomer

Dominant peak at 9.8 mL (Very Large (HMW) zone), 85% of total area, FWHM = 1.30 mL. Secondary peak(s) at 15.0 mL (15%). Elution in the HMW zone is consistent with higher-order oligomeric assembly.

Recommendation: Priority: cryo-EM / native MS

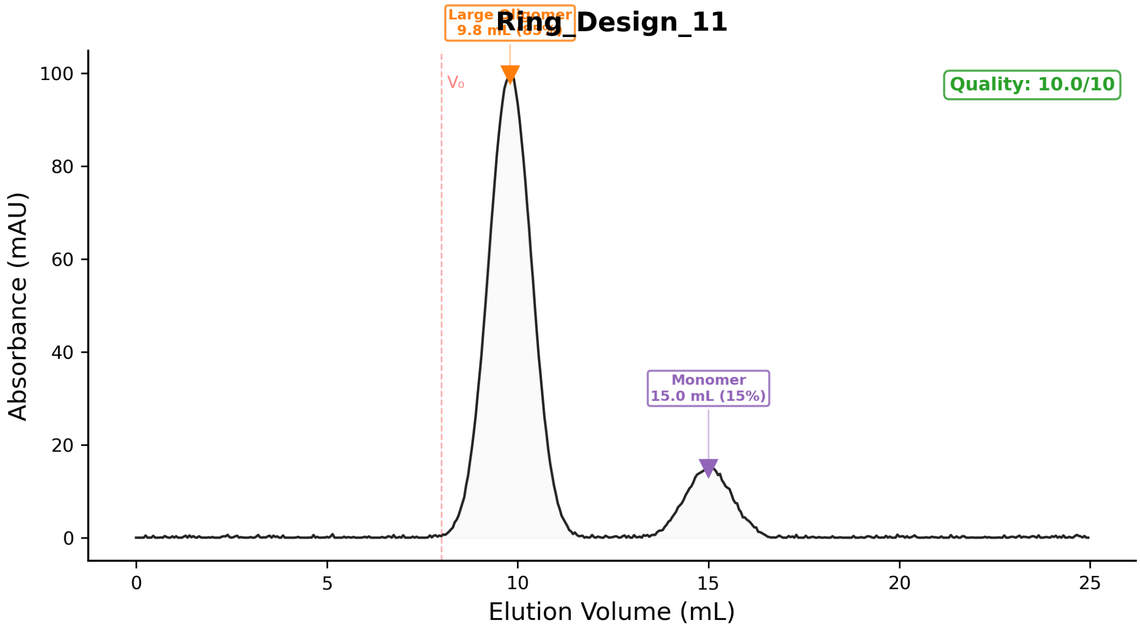

Figure 5: Annotated SEC chromatogram — Ring_Design_11.

Dimer_Variant_03

#	VE (ML)	HEIGHT (MAU)	FWHM (ML)	AREA %	ZONE
1	10.50	24.7	1.06	18.7%	Very Large (HMW)
2	13.50	80.0	1.65	81.3%	Mid-size

Dominant peak at 13.5 mL (Mid-size zone), 81% of total area, FWHM = 1.65 mL. Secondary peak(s) at 10.5 mL (19%).

Recommendation: Use as positive control

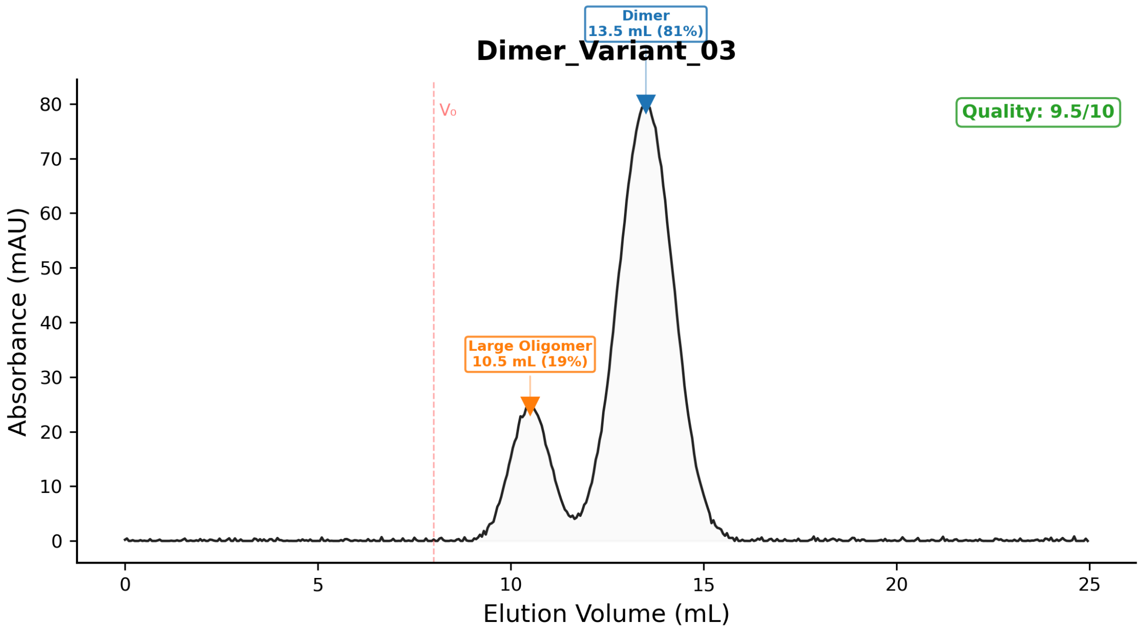

Figure 6: Annotated SEC chromatogram — Dimer_Variant_03.

Mixed_Assembly_07

#	VE (ML)	HEIGHT (MAU)	FWHM (ML)	AREA %	ZONE
1	9.55	47.5	0.57	24.8%	Very Large (HMW)
2	11.00	51.2	4.70	31.4%	Large Oligomer
3	13.00	40.3	0.92	27.4%	Mid-size
4	15.50	30.0	0.98	16.3%	Small / Monomer

Dominant peak at 11.0 mL (Large Oligomer zone), 31% of total area, FWHM = 4.70 mL. Secondary peak(s) at 9.6 mL (25%), 13.0 mL (27%), 15.5 mL (16%). Polydisperse — multiple species present. Broad peak(s) suggest conformational heterogeneity or equilibrium between oligomeric states. Elution in the HMW zone is consistent with higher-order oligomeric assembly.

Recommendation: Buffer optimisation

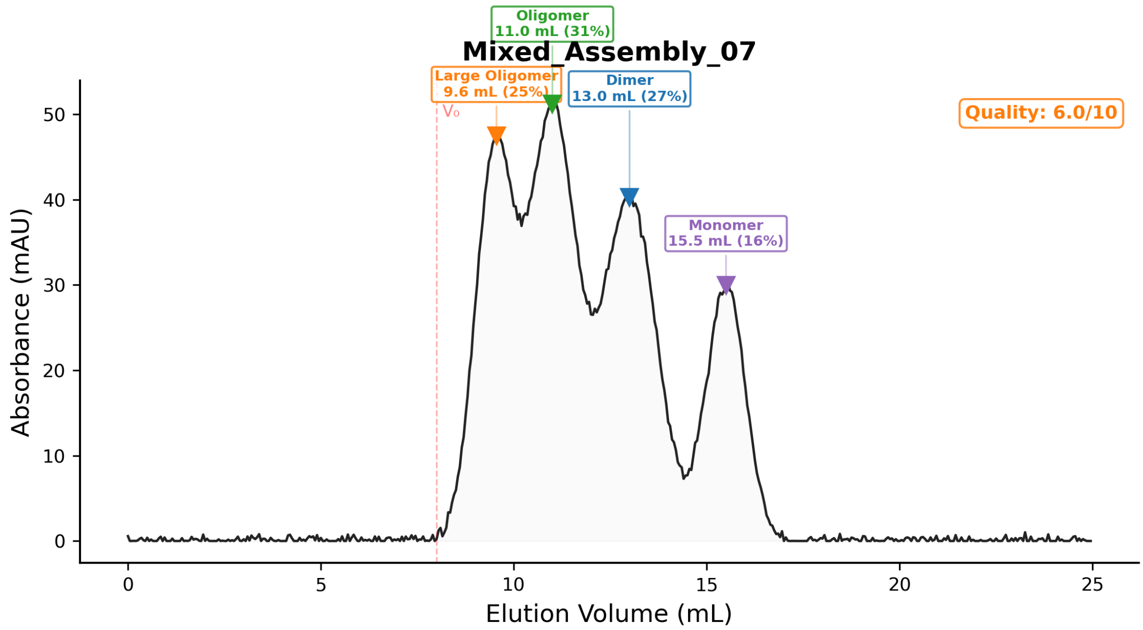

Figure 7: Annotated SEC chromatogram — Mixed_Assembly_07.

Aggregator_05

#	VE (ML)	HEIGHT (MAU)	FWHM (ML)	AREA %	ZONE
1	8.00	89.2	0.96	50.1%	Void / Aggregate
2	11.45	30.1	1.83	32.6%	Large Oligomer
3	15.00	20.0	1.31	17.3%	Small / Monomer

Dominant peak at 8.0 mL (Void / Aggregate zone), 50% of total area, FWHM = 0.96 mL. Secondary peak(s) at 11.4 mL (33%), 15.0 mL (17%). Significant void-volume signal (50%) indicates non-specific aggregation. Polydisperse — multiple species present.

Recommendation: Redesign interface

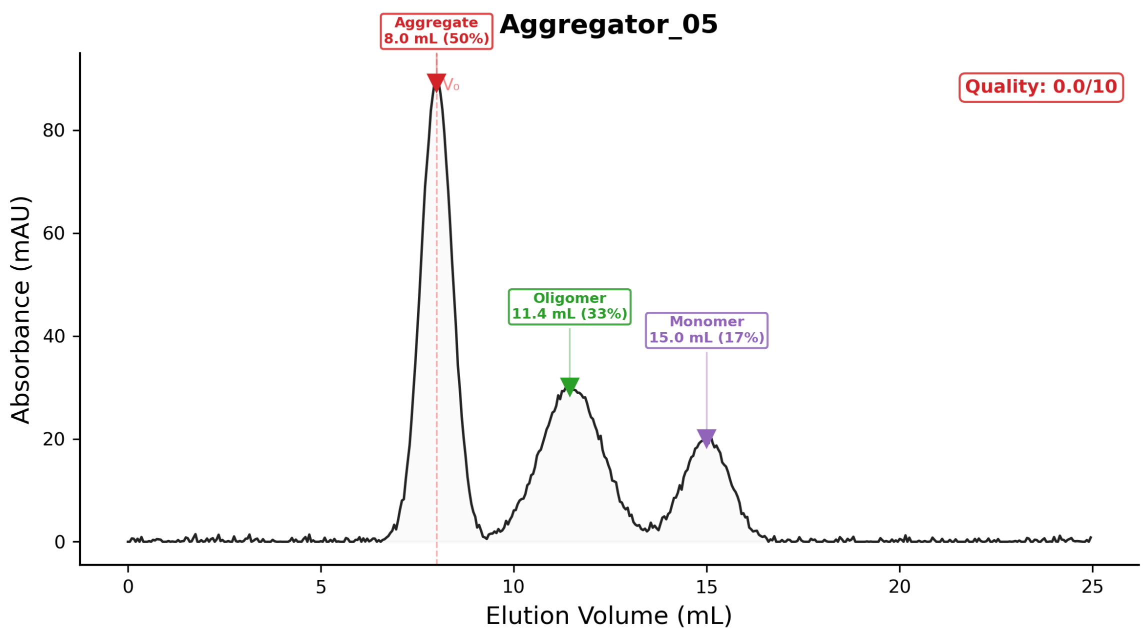

Figure 8: Annotated SEC chromatogram — Aggregator_05.

Note
This appendix contains all per-construct data. See the main report for summary, discussion, and recommendations.